SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA37- SOFTWARE TESTING

EXP:9

Name: Balaguru S

Register Number: 192312070

Requirement Analysis and Negative Test Scenarios and Test Cases for a Banking Application

Aim

To study the requirements of a Banking Application, and to prepare negative test scenarios and test cases that confirm the application handles invalid inputs and wrong user actions safely.

Problem Statement

An internet banking application lets customers check balances, transfer money, pay bills and download statements. It handles money and confidential data, so a defect can cause financial loss, security breaches or legal problems for the bank.

Testing is needed to check that the application:

- Customers can log in securely and complete banking transactions correctly.
- Invalid inputs and unauthorized actions are rejected safely.
- Limits, balance checks and security rules follow the requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	The customer logs in with a user ID and password, followed by an OTP (two-factor authentication).
FR2	The account is locked after 3 consecutive wrong password attempts.
FR3	The customer can view the account balance and a mini statement of the last 5 transactions.
FR4	The customer can add a beneficiary with name, account number and a valid IFSC code, verified by OTP.
FR5	Fund transfer is allowed only to an added beneficiary. The minimum is ₹1 and the daily limit is ₹1,00,000.
FR6	A transfer must not exceed the available balance.
FR7	Every transaction gives a reference number, a receipt and an SMS alert.
FR8	The customer can pay utility bills and recharge mobile numbers.
FR9	The customer can download a statement for a chosen period as a PDF. The from date must not be after the to date.
FR10	The customer can change the password. The new password must differ from the last 3 passwords and be strong.
FR11	The session ends after 5 minutes of inactivity, and the customer can log out securely.

ID	Requirement
FR12	Only logged-in customers can access account pages.

2 Software and environment requirements

Item	Details
Application	Web banking portal (Java Spring Boot back end, Angular front end)
Database	Oracle / MySQL (accounts, beneficiaries, transactions)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, Postman, bug tracker (Jira)
Test data	Savings and current accounts with different balances, valid and invalid IFSC codes, beneficiaries, sandbox OTP service

Negative Test Scenarios

A negative test scenario checks that the system rejects invalid input or a wrong action, shows a clear message and stays stable and secure.

Scenario ID	Module	Test Scenario	Requirement
BK-NS01	Login	Log in with a wrong password	FR1
BK-NS02	Login	Account lock after 3 wrong attempts	FR2
BK-NS03	Login	Enter a wrong OTP	FR1
BK-NS04	Login	Submit blank user ID and password	FR1
BK-NS05	Beneficiary	Add a beneficiary with an invalid IFSC	FR4
BK-NS06	Beneficiary	Add a beneficiary with an invalid account number	FR4
BK-NS07	Transfer	Transfer more than the available balance	FR6
BK-NS08	Transfer	Transfer above the daily limit	FR5
BK-NS09	Transfer	Enter an invalid amount	FR5
BK-NS10	Transfer	Transfer to a beneficiary that was not added	FR5
BK-NS11	Statement	Download a statement with invalid dates	FR9
BK-NS12	Password	Change to the old or a weak password	FR10
BK-NS13	Session	Use the portal after the session timeout	FR11
BK-NS14	Access control	Open an account page without logging in	FR12

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
BK-NT01	BK-NS01	1. Enter a valid user ID and a wrong password 2. Click Login	100234; Wrong@123	Login fails with “Invalid credentials”. The OTP is not sent.
BK-NT02	BK-NS02	1. Enter a wrong password three times in a row 2. Then enter the correct password	Wrong@1, Wrong@2, Wrong@3, then Bank@2026	Account is locked after the third attempt. Even the correct password is rejected.
BK-NT03	BK-NS03	1. Enter valid credentials 2. Enter a wrong OTP	654321 (wrong)	Login fails with “Invalid OTP” and the dashboard is not shown.
BK-NT04	BK-NS04	1. Leave both fields empty 2. Click Login	Blank	Mandatory-field messages are displayed.
BK-NT05	BK-NS05	1. Enter name and account number 2. Enter an invalid IFSC 3. Submit	IFSC: ABCD123	Error “Invalid IFSC code” is shown and the beneficiary is not added.
BK-NT06	BK-NS06	1. Enter an account number with letters or too few digits 2. Submit	12AB34; 123	Validation error is shown and the beneficiary is not added.
BK-NT07	BK-NS07	1. Select a beneficiary 2. Enter an amount above the balance 3. Confirm	₹30,000 with balance ₹25,000	Transfer is rejected with “Insufficient balance”. The balance is unchanged.
BK-NT08	BK-NS08	1. Select a beneficiary 2. Enter an amount above the limit 3. Confirm	₹1,00,001	Transfer is rejected with “Daily limit exceeded”.
BK-NT09	BK-NS09	1. Enter 0, a negative number and letters as the amount	0; -500; abc	Each entry is rejected with a validation message.
BK-NT10	BK-NS10	1. Try to transfer directly to an account number 2. Skip the add beneficiary step	Account 999988887777	The transfer is not allowed until the beneficiary is added and verified.
BK-NT11	BK-NS11	1. Select a from date later than the to date 2. Click Download	From 31-10-2026; To 01-10-2026	Error “From date cannot be after To date” is shown. No file is generated.
BK-NT12	BK-NS12	1. Enter the current password as the new password 2. Then enter a weak password	Old: Bank@2026; New: Bank@2026, then 1234	Both attempts are rejected with a clear message.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
BK-NT13	BK-NS13	1. Log in 2. Stay idle for more than 5 minutes 3. Click any menu	Idle for 6 minutes	Session has expired and the user is taken to the login page.
BK-NT14	BK-NS14	1. Copy the account summary URL 2. Open it in a new private window without logging in	Direct URL	User is redirected to the login page and no data is shown.

Result / Conclusion

Result: 14 negative test scenarios and 14 test cases were prepared for the Banking Application, covering 9 of the 12 functional requirements (no negative scenario for FR3, FR7, FR8). The test cases by module are:

Module	Test Scenarios	Test Cases
Login	4	4
Beneficiary	2	2
Transfer	4	4
Statement	1	1
Password	1	1
Session	1	1
Access control	1	1
Total	14	14

Conclusion: The negative test cases cover invalid inputs, boundary violations, failed payments or operations, and unauthorized actions in the Banking Application. When they are executed on a build, they confirm that the system rejects wrong input safely, shows clear messages, and keeps the data and the user’s session secure. Together with the positive test cases, they give a complete set of tests for the stated requirements.